

Northwind Data Systems
Ingestion Reliability Review

This review covers the twelve months to March 2024. It summarises the incidents recorded against the ingestion service, the changes made in response, and the residual risks the team has accepted rather than mitigated.

Three incidents in the period were classed as data loss. In each case the loss was silent: the pipeline reported success and the absence of data was noticed by a human reading a dashboard rather than by any automated check.

The common cause was the same in all three. Each stage validated that it had completed, and no stage validated that it had produced a plausible quantity of output. A job that processes zero records successfully processes zero records.

Changes made

Row-count assertions were added at the manifest boundary and at the load boundary. Both compare against a trailing median rather than a fixed floor, because a fixed floor is wrong for every table except the one it was set for.

Extraction-yield gates were added to the parsing stage. A document producing fewer than one hundred characters per page is now routed to the OCR queue instead of being indexed as an empty document.

Residual risks

The team has accepted that documents in the quarantine bucket are drained manually and that no service level objective covers quarantine age.